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Teo aut disco unidad composicion

Teorema 1. *Sean* $f = \rho_{\gamma_1} \circ \tau_{a_1}$ *y* $g = \rho_{\gamma_2} \circ \tau_{a_2}$ *en* $\operatorname{Aut}(\mathbb{D})$, *entonces*

$$g \circ f = \rho_\gamma \circ \tau_a \quad \textit{donde} \quad a = \tau_{a_1}(a_2) = \frac{a_1 - a_2}{1 - \overline{a_1} a_2} \quad \wedge \quad \gamma = \begin{cases} \gamma_1 \, \tau_{\overline{a_1} a_2}(\gamma_2) & \textit{si } a_2 \neq a_1 \gamma_1, \\ -\gamma_1 \gamma_2 & \textit{si } a_2 = a_1 \gamma_1, \end{cases}$$

En particular, si $a_2 = a_1 \gamma_1$, *entonces* $g \circ f = \rho_{\gamma_1 \gamma_2}$.

***Demostración*:** Operando: $g \circ f = \rho_{\gamma_2} \circ (\tau_{a_2} \circ \rho_{\gamma_1}) \circ \tau_{a_1} \overset{\substack{\texttt{lem-involucion-disco-unidad-rho/1} \\ \downarrow}}{=} \rho_{\gamma_2} \circ \rho_{\gamma_1} \circ (\tau_{\overline{\gamma_1} a_2} \circ \tau_{a_1})$.

Ahora bien, por el `lem-aut-disco-unidad-composicion-involuciones/`lema 1, tenemos que $\tau_{\overline{\gamma_1} a_2} \circ \tau_{a_1} = \rho_{\widetilde{\gamma}} \circ \tau_{\widetilde{a}}$ donde

$$\widetilde{\gamma} = \begin{cases} \tau_{\overline{a_1} a_2 \overline{\gamma_2}}(1) & \text{si } a_1 \neq \overline{\gamma_1} a_2, \\ -1 & \text{si } a_1 = \overline{\gamma_1} a_2, \end{cases} \quad \wedge \quad \widetilde{a} = \tau_{a_1}(\overline{\gamma_1} a_2) = \frac{a_1 - \overline{\gamma_1} a_2}{1 - \overline{a_1} \overline{\gamma_1} a_2}.$$

Concluimos por tanto que si $g \circ f = \rho_\gamma \circ \tau_a$, entonces

$$\gamma = \gamma_1 \gamma_2 \widetilde{\gamma} = \begin{cases} \gamma_1 \gamma_2 \cdot \tau_{\overline{a_1} a_2 \overline{\gamma_2}}(1) & \text{si } a_2 \neq a_1 \gamma_1, \\ -\gamma_1 \gamma_2 & \text{si } a_2 = a_1 \gamma_1, \end{cases} \quad \wedge \quad a = \widetilde{a} = \tau_{a_1}(\overline{\gamma_1} a_2).$$

Finalmente, tenemos que $\gamma_2 \cdot \tau_{\overline{a_1} a_2 \overline{\gamma_2}}(1) = \gamma_2 \cdot \dfrac{\overline{a_1} a_2 \overline{\gamma_2} - 1}{1 - a_1 \overline{a_2} \gamma_2} = \dfrac{\overline{a_1} a_2 - \gamma_2}{1 - a_1 \overline{a_2} \gamma_2} = \tau_{\overline{a_1} a_2}(\gamma_2)$ y se sigue el resultado deseado. ■